

Integration WS 4

Evaluate Q1-10

1. $\int \sin 4x \cos 3x \, dx$

2. $\int \cos 5x \sin 2x \, dx$

3. $\int_0^{\frac{\pi}{4}} (\tan^3 x + \tan x) \, dx$

4. $\int_0^{\frac{\pi}{3}} (\sin^3 x - \sin x) \, dx$

5. $\int_0^{\frac{\pi}{2}} \sqrt{\cos x} \sin^3 x \, dx$

6. $\int_0^{\frac{\pi}{4}} \sin 4x \sin 2x \, dx$

7. $\int_0^1 x e^{2x} \, dx$

8. $\int_1^e (\ln x)^2 \, dx$

9. $\int_0^{\frac{\pi}{2}} x \cos x \, dx$

10. $\int_0^{\frac{\pi}{2}} x \sin x \cos x \, dx$

11. If $I_n = \int \tan^n x \, dx$ for $n \geq 0$ show that

$$I_n = \frac{1}{n-1} \tan^{n-1} x - I_{n-2} \text{ for } n \geq 2$$

12. If $I_n = \int x(\ln x)^n \, dx$ for $n \geq 0$ show that

$$I_n = \frac{1}{2} x^2 (\ln x)^n - \frac{n}{2} I_{n-1} \text{ for } n \geq 1$$

13. If $I_n = \int \sin^n x \, dx$ for $n \geq 0$ show that

$$I_n = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} I_{n-2} \text{ for } n \geq 2.$$

Hence show that if $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ for

$n \geq 0$, then $I_n = \frac{n-1}{n} I_{n-2}$ for $n \geq 2$ and

deduce that $I_5 \cdot I_6 = \frac{\pi}{12}$

14. If $I_n = \int \sec^n x \, dx$ for $n \geq 0$ show that

$$I_n = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2} \text{ for } n \geq 2.$$

Hence show that if $I_n = \int_0^{\frac{\pi}{4}} \sec^n x \, dx$ for

$n \geq 0$, then $I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{n-2}{(n-1)} I_{n-2}$ for $n \geq 2$

and deduce that $I_6 = \frac{28}{15}$.

15. If $I_n = \int_0^{\frac{\pi}{2}} x^n \cos x \, dx$ for $n \geq 0$ show that

$$I_n = \left(\frac{\pi}{2}\right)^n - n(n-1)I_{n-2} \text{ for } n \geq 2. \text{ Hence evaluate } I_6.$$

16. If $I_n = \int_0^1 x(1-x^3)^n \, dx$ for $n \geq 0$ show that

$$I_n = \frac{3n}{3n+2} I_{n-1} \text{ for } n \geq 1. \text{ Hence find an expression for } I_n \text{ in terms of } n \text{ for } n \geq 0.$$

17. a. Find the rational numbers A, B and C such

$$\text{that } \frac{x-1}{x^3+1} = \frac{A}{x+1} + \frac{Bx+C}{x^2-x+1}.$$

b. Hence show that $\int_0^1 \frac{x^3+x}{x^3+1} \, dx = 1 - \frac{2}{3} \ln 2$

18. Use integration by parts to evaluate

$$I = \int_0^{\frac{\pi}{4}} \sec^3 x \, dx$$

19. In each case let $t = \tan \frac{x}{2}$ in order to show

that:

a. $\int_0^{\frac{\pi}{2}} \frac{1}{3+5 \cos x} \, dx = \frac{1}{4} \ln 3$

b. $\int_0^{\frac{\pi}{2}} \frac{1}{\cos x - 2 \sin x + 3} \, dx = \frac{\pi}{4}$

20. a. Find the values of A, B, C and D such that

$$\frac{4t}{(1+t)^2(1+t^2)} = \frac{A}{1+t} + \frac{B}{(1+t)^2} + \frac{Ct+D}{1+t^2}$$

b. Hence use the t -substitution to evaluate

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{1+\sin x} \, dx$$

21. Use the substitution $u = \sqrt[6]{x}$ to show that

$$\int_1^{64} \frac{1}{\sqrt{x} + \sqrt[3]{x}} \, dx = 11 - 6 \ln \left(\frac{3}{2}\right)$$

22. Find $\int \sqrt{a^2 - x^2} \, dx$ using

a. The substitution $\theta = \sin^{-1} \frac{x}{a}$

b. Integration by parts

23. a. Show that $\int_0^1 \frac{5-5x^2}{(1+2x)(1+x^2)} \, dx =$

$$\frac{1}{2} \left(\pi + \ln \frac{27}{16} \right)$$

b. Hence find $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\cos x + 2 \sin x} \, dx$ using the substitution $t = \tan \frac{x}{2}$

24. a. Find integers P and Q such that $8 \sin x + \cos x - 2 = P(3 \sin x + 2 \cos x - 1) + Q(3 \cos x - 2 \sin x)$

b. Hence find $\int \frac{8 \sin x + \cos x - 2}{3 \sin x + 2 \cos x - 1} dx$

25. a. If $T_n = \int_0^\pi \sin^n x dx$, show that $T_n =$

$$\frac{n-1}{n} T_{n-2}$$

b. Hence show that $T_5 T_6 = \frac{\pi}{3}$

26. a. Show that

$$\cos(ax) \sin(bx) = \frac{\sin[(a+b)x] - \sin[(a-b)x]}{2}$$

b. Hence, or otherwise, find

$$\int \cos(ax) \sin(bx) dx$$

c. If $a = b$ in the formula in i. the formula

$$\text{becomes } \cos(ax) \sin(ax) = \frac{\sin(2ax)}{2}$$

i. Use this new formula to find

$$\int \cos(ax) \sin(ax) dx$$

ii. Find $\int \cos(ax) \sin(ax) dx$ without using this formula (ie. An alternative way)

iii. The very good mathematician David

Hattelton is confused. He successfully

complete parts (i) and (ii) but noticed

his answers looked very different.

Show, by manipulation, that his two

answers are in fact the same.

27. Let $I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^n x dx$ where n is a positive

integer

a. Using integration by parts, show that

$$I_n = \frac{1}{n-1} (2^{n-2} \sqrt{3} + (n-2) I_{n-2})$$

b. Hence, or otherwise, evaluate

$$\int_0^{\frac{\pi}{3}} \sec^4 x dx$$

28. If $I_n = \int_0^1 (1 - \sqrt{x})^n dx$ for $n \geq 0$, show that

$$I_n = \frac{n}{n+2} I_{n-1} \text{ for } n \geq 1 \text{ and hence evaluate}$$

$$I_{100}.$$

29. a. Show that $\frac{t^n}{1+t^2} = t^{n-2} - \frac{t^{n-2}}{1+t^2}$

b. Let $I_n = \int \frac{t^n}{1+t^2} dt$, show that $I_n = \frac{t^{n-1}}{n-1} -$

$$I_{n-2}, n \geq 2$$

c. Show that $\int_0^1 \frac{t^6}{1+t^2} dt = \frac{13}{15} - \frac{\pi}{4}$

30. Find $\int \frac{4x-3}{\sqrt{6+2x-3x^2}} dx$

31. Use integration by parts to show that

$$\int x^3 e^{-x^2} dx = -\frac{1}{2} e^{-x^2} (1 + x^2) + C$$

a. $I_n = \int_0^a (a-x)^n \cos x dx$, $a > 0$ and n is an integer with $n \geq 0$.

b. Show that, for $n \geq 2$, $I_n = na^{n-1} - n(n-1)I_{n-2}$.

c. Hence evaluate $\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - x\right)^3 \cos x dx$

Answer

1. $-\frac{1}{14} \cos 7x - \frac{1}{2} \cos x$

2. $\frac{1}{6} \cos 3x - \frac{1}{14} \cos 7x$

3. $\frac{1}{2}$

4. $-\frac{7}{24}$

5. $\frac{8}{21}$

6. $\frac{1}{3}$

7. $\frac{1}{4} (e^2 + 1)$

8. $e - 2$

9. $\frac{\pi}{2} - 1$

10. $\frac{\pi}{8}$

15. $\left(\frac{\pi}{2}\right)^6 - 30 \left(\frac{\pi}{2}\right)^4 + 360 \left(\frac{\pi}{2}\right)^2 - 720$

16. $\frac{3^n n!}{(3n+2)(3n-1) \dots 8.5.2}$

17. a. $A = -\frac{2}{3}, B = \frac{2}{3}, C = -\frac{1}{3}$

18. $\frac{1}{\sqrt{2}} + \frac{1}{2} \ln(1 + \sqrt{2})$

20. a. $A = 0, B = -2, C = 0, D = 2$; b.

$$\frac{\pi}{2} - 1$$

22. $\frac{1}{2} a^2 \sin^{-1} \frac{x}{a} + \frac{1}{2} x \sqrt{a^2 - x^2} + C$

23. b. $\frac{1}{10} \left(\pi + \ln \frac{27}{16} \right)$

24. a. $P = 2, Q = -1$; b. $2x -$

$$\ln(3 \sin x + 2 \cos x - 1) + C$$

26. b. $\frac{\cos[(a-b)x]}{2(a-b)} - \frac{\cos[(a+b)x]}{2(a+b)} + C$; c. i.

$$-\frac{\cos(2ax)}{4a} + C, \text{ ii. } \frac{1}{a} \left(\frac{\sin^2 ax}{2} \right) + C$$

27. b. $2\sqrt{3}$

28. $I_{100} = \frac{2}{102 \times 101}$

30. $-\frac{4}{3} \sqrt{6+2x-3x^2} -$

$$\frac{5}{3\sqrt{3}} \sin^{-1} \left(\frac{3x-1}{\sqrt{19}} \right) + C$$

31. $\frac{3\pi^2}{4} - 6$